

EX : 5 Create an Android application to display X, Y sensor values using the Sensor Service.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Accelerometer Sensor"
        android:textStyle="bold"
        android:textSize="24sp"/>

    <TextView
        android:id="@+id/tvX"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="X : 0.0"
        android:textSize="20sp"
        android:layout_marginTop="20dp"/>

    <TextView
        android:id="@+id/tvY"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Y : 0.0"
        android:textSize="20sp"
        android:layout_marginTop="10dp"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.sensordemo

import android.content.Context
import android.hardware.Sensor
import android.hardware.SensorEvent
import android.hardware.SensorEventListener
import android.hardware.SensorManager
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.TextView

class MainActivity : AppCompatActivity(), SensorEventListener {
    private lateinit var sensorManager: SensorManager

    private var sensor: Sensor? = null

    private lateinit var tvX: TextView
    private lateinit var tvY: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
        tvX = findViewById(R.id.tvX)
        tvY = findViewById(R.id.tvY)
    }
}
```

```

sensorManager = getSystemService(Context.SENSOR_SERVICE) as
SensorManager

    sensor =
sensorManager.getDefaultSensor(Sensor.TYPE_ACCELEROMETER)

}

override fun onResume() {

    super.onResume()

    sensor?.also {

        sensorManager.registerListener(this, it,
SensorManager.SENSOR_DELAY_NORMAL)

    }

}

override fun onPause() {

    super.onPause()

    sensorManager.unregisterListener(this)

}

override fun onSensorChanged(event: SensorEvent?) {

    if (event != null) {

        tvX.text = "X : ${event.values[0]}"

        tvY.text = "Y : ${event.values[1]}"

    }

}

override fun onAccuracyChanged(sensor: Sensor?, accuracy: Int) {

}

}

```

OUTPUT

